

EXP 2: For a given set of training data examples stored in a .CSV file, implement and demonstrate the Candidate-Elimination algorithm in python to output a description of the set of all hypotheses consistent with the training examples

Aim

To implement and demonstrate the **Candidate Elimination Algorithm** using a training dataset stored in a CSV file and determine the set of all hypotheses that are consistent with the given training examples.

Algorithm Steps

1. Initialize the specific hypothesis (S) using the first positive example.
2. Initialize the general hypothesis (G) with the most general hypothesis.
3. For each training example:
 - If the example is **positive**:
 - Generalize S if necessary.
 - Remove hypotheses from G that are inconsistent.
 - If the example is **negative**:
 - Specialize G to exclude the negative example.
 - Remove inconsistent hypotheses.
4. The remaining hypotheses between S and G form the version space.

Program:

```
import pandas as pd

data = pd.read_csv("trainingdata.csv")

print("Training Data:\n")

print(data)

concepts = data.iloc[:, :-1].values

target = data.iloc[:, -1].values

def initialize_G(num_attributes):

    return [['?' for _ in range(num_attributes)]]

def initialize_S(num_attributes):

    return ['0' for _ in range(num_attributes)]
```

```

def candidate_elimination(concepts, target):
    num_attributes = len(concepts[0])
    S = initialize_S(num_attributes)
    G = initialize_G(num_attributes)
    print("\nInitial Specific Hypothesis (S):")
    print(S)
    print("\nInitial General Hypothesis (G):")
    print(G)
    for i, instance in enumerate(concepts):
        if target[i] == "Yes":
            for j in range(num_attributes):
                if S[j] == '0':
                    S[j] = instance[j]
                elif S[j] != instance[j]:
                    S[j] = '?'
            G = [g for g in G if all(
                g[j] == '?' or g[j] == instance[j]
                for j in range(num_attributes)
            )]
        else:
            G_new = []
            for g in G:
                for j in range(num_attributes):
                    if g[j] == '?':
                        if S[j] != '?':
                            new_hypothesis = g.copy()
                            new_hypothesis[j] = S[j]
                            G_new.append(new_hypothesis)
            G = G_new
    print(f"\nAfter Training Example {i+1}:")

```

```

    print("S =", S)

    print("G =", G)

    return S, G

S_final, G_final = candidate_elimination(concepts, target)

print("\n=====")

print("Final Specific Hypothesis (S)")

print(S_final)

print("\nFinal General Hypothesis (G)")

for hypothesis in G_final:
    print(hypothesis)

```

Output:

```

--- Training Data:

  Sky AirTemp Humidity Wind Water Forecast EnjoySport
0 Sunny Warm Normal Strong Warm Same Yes
1 Sunny Warm High Strong Warm Same Yes
2 Rainy Cold High Strong Warm Change No
3 Sunny Warm High Strong Cool Same Yes

Initial Specific Hypothesis (S):
['?', '?', '?', '?', '?', '?', '?']

Initial General Hypothesis (G):
[['?', '?', '?', '?', '?', '?', '?']]

After Training Example 1:
S = ['Sunny', 'Warm', 'Normal', 'Strong', 'Warm', 'Same']
G = [['?', '?', '?', '?', '?', '?', '?']]

After Training Example 2:
S = ['Sunny', 'Warm', '?', 'Strong', 'Warm', 'Same']
G = [['?', '?', '?', '?', '?', '?', '?']]

After Training Example 3:
S = ['Sunny', 'Warm', '?', 'Strong', 'Warm', 'Same']
G = [['Sunny', '?', '?', '?', '?', '?'], ['?', 'Warm', '?', '?', '?', '?'], ['?', '?', '?', 'Strong', '?', '?'], ['?', '?', '?', '?', 'Warm', '?'], ['?', '?', '?', '?', '?', 'Same']]

After Training Example 4:
S = ['Sunny', 'Warm', '?', 'Strong', '?', 'Same']
G = [['Sunny', '?', '?', '?', '?', '?'], ['?', 'Warm', '?', '?', '?', '?'], ['?', '?', '?', 'Strong', '?', '?'], ['?', '?', '?', '?', '?', 'Same']]

=====
Final Specific Hypothesis (S)
['Sunny', 'Warm', '?', 'Strong', '?', 'Same']

Final General Hypothesis (G)
['Sunny', '?', '?', '?', '?', '?']
['?', 'Warm', '?', '?', '?', '?']
['?', '?', '?', 'Strong', '?', '?']
['?', '?', '?', '?', '?', 'Same']

```

Result:

The Candidate Elimination Algorithm was successfully implemented in Python using a CSV dataset.